



POLITECNICO

MILANO 1863

Financial Engineering

Assignment 4 Group 1

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1. Case study: Certificate Pricing

This exercise focuses on the pricing and structuring of an Equity Protection Certificate issued by Bank XX on February 15th, 2008. The primary objective is to determine the fair upfront fee ($X\%$) to be paid by the Investment Bank at inception under a single-curve interest rate modeling framework, assuming zero counterparty risk due to an ISDA agreement with a Credit Support Annex (CSA).

The certificate has a principal amount of 100€ million and a maturity of 5 years. It guarantees a protection level of 95% of the principal and pays an exotic structured coupon at maturity, which consists of a 110% participation in the positive performance (above the protection level) of an equally weighted arithmetic basket of two equities: ENI S.p.A. and AXA S.A.

a. Methodology

The valuation of the Equity Protection Certificate relies on an Asset Liability Management (ALM) framework, where the issuer (Party A) hedges the exotic risk of the product by entering into a swap agreement with the Investment Bank (Party B).

Structuring and Swap Valuation

To eliminate market exposure, the swap is structured such that the Net Present Value (NPV) of the legs exchanged between Party A (Bank XX) and Party B (Investment Bank) must be equal at the start date t_0 .

Party A pays a floating rate leg (Euribor 3m + spread s) and a maturity payment equal to $(1 - P)$ of the Principal amount N , where P is the protection level. Party B pays the upfront percentage $X\%$ at inception and the exotic coupon at maturity t_N .

By equating the two NPVs, the upfront percentage is isolated as:

$$X\% = 1 - B(t_0, t_N) + s \cdot BPV + (1 - P)B(t_0, t_N) - c \quad (1)$$

where $c = C/N$ is the normalized price of the option component, $B(t_0, t_N)$ is the discount factor at maturity, BPV is the Basis Point Value of the floating leg, computed over the adjusted payment dates using the *Act/360* day-count convention.

Exotic Coupon Pricing

The coupon is a European Call option on an equally weighted arithmetic basket of two assets (ENI and AXA), with a participation coefficient α . The payoff at maturity is:

$$\text{Payoff} = N \cdot \alpha \cdot \max(\text{Basket}_{t_N} - P, 0) \quad (2)$$

Because the sum of lognormally distributed variables is not strictly lognormal, a closed-form analytical solution does not exist. The pricing is therefore performed using two alternative numerical methods: Moment Matching and Monte Carlo simulation.

Approach 1: Moment Matching (Lognormal Approximation)

This method approximates the distribution of the arithmetic basket with a synthetic lognormal distribution by matching the first two moments. Let $\tilde{E}_t^i = E_t^i / E_0^i$ represent the normalized asset prices. Their expected forward values are $F_i = \exp((r - q_i)T)$, where r is the continuous risk-free rate and q_i the continuous dividend yield.

The first moment (expected value M_1) and the second moment (M_2) of the basket are computed analytically and, by assuming the synthetic basket B_T follows a lognormal distribution $\text{LogN}(m, \Sigma^2 T)$, its raw moments satisfy $\mathbb{E}[B_T^k] = \exp(km + \frac{1}{2}k^2\Sigma^2 T)$.

It follows that:

$$\frac{M_2}{M_1^2} = \frac{\exp(2m + 2\Sigma^2 T)}{\exp(2m + \Sigma^2 T)} = \exp(\Sigma^2 T) \implies \Sigma^2 = \frac{1}{T} \ln \left(\frac{M_2}{M_1^2} \right) \quad (3)$$

The option price is finally evaluated using the standard Black-76 formula, treating M_1 as the forward price, Σ as the volatility, and the protection level P as the strike.

Approach 2: Monte Carlo Simulation

To validate the lognormal approximation, the basket is also priced via Monte Carlo simulation. The correlated asset paths are simulated under the risk-neutral measure using the exact solution to the Geometric Brownian Motion (GBM) Stochastic Differential Equation.

To account for the correlation ρ between the two assets, independent standard normal variables $Z_1, Z_2 \sim \mathcal{N}(0, 1)$ are drawn and transformed via Cholesky decomposition:

$$W_1 = Z_1 \quad (4)$$

$$W_2 = \rho Z_1 + \sqrt{1 - \rho^2} Z_2 \quad (5)$$

The terminal normalized prices at maturity T are simulated as:

$$\tilde{E}_T^i = \exp \left(\left(r - q_i - \frac{1}{2} \sigma_i^2 \right) T + \sigma_i \sqrt{T} W_i \right) \quad (6)$$

The basket value is reconstructed as $Basket_T = w_1 \tilde{E}_T^1 + w_2 \tilde{E}_T^2$, and the Call price is obtained as the discounted arithmetic average of the simulated payoffs over $N_{sim} = 10^7$ iterations.

b. Results and Discussion

The following table summarizes the valuation of the structured coupon and the resulting upfront fee under the two different pricing methodologies.

Methodology	Coupon Value (EUR)	Final Upfront (X%)
Moment Matching (Analytical)	18,120,357.15	9.9586 %
Monte Carlo Simulation	18,110,828.23	9.9682 %
Absolute Difference	9,528.92	0.0095 %

Table 1: Comparison of the structured coupon value and resulting upfront fee: Moment Matching vs. Monte Carlo.

As reported in the Table above, the absolute difference between the analytical approximation and the simulated coupon is roughly 9,528.92 €. While this figure may appear significant in isolation, it corresponds to a relative discrepancy of less than 0.01% with respect to the 100,000,000 € notional amount. The Monte Carlo simulation yielded a standard error of 8.2×10^{-5} on the unit price, confirming that the simulation noise is well-controlled and that the observed difference represents a structural discrepancy between the true basket distribution and the synthetic lognormal approximation, even if the tight convergence between the two methods confirms that the Moment Matching approach is highly robust for evaluating the exotic leg of this certificate, providing near-simulation accuracy.

Finally, the slight overestimation of the coupon value by the Moment Matching approach directly translates into a lower upfront fee. The analytical model requires Party B to pay an upfront of 9,958,640.66 €, whereas the Monte Carlo model evaluates the fair upfront at 9,968,169.58 €.

2. Exercise 2: Pricing Digital Option

This exercise investigates the pricing of a cash-or-nothing digital call option on the Eurostoxx 50 index, with the following contractual specifications: notional of 10€ million, digital payoff of 5% of notional, expiry of 1 year (Act/365), and discount curve as of February 15th, 2008.

The objective is to quantify the pricing discrepancy arising from two alternative approaches: the standard **Black (1976)** model, which assumes a flat implied volatility surface, and the **implied volatility smile** approach, which accounts for the market-observed skew in the volatility surface of the underlying.

a. Methodology

Under the Black framework, the cash-or-nothing digital call price is obtained as the negative derivative of the vanilla call price with respect to the strike:

$$dc_B(K) = -\frac{\partial c_B(K)}{\partial K} = B(t_0, t)\mathcal{N}(d_2), \quad (7)$$

where

$$d_2 = \frac{\ln(F_0/K) - \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}}. \quad (8)$$

The forward price is computed as

$$F_0 = S_0 \frac{e^{-d\tau}}{B(t_0, t)}, \quad (9)$$

and the ATM volatility σ is obtained by cubic spline interpolation of the market smile at $K = F_0$. Since the Black model assumes a flat volatility surface, the market smile is incorporated by interpreting the digital as the limit of a bull spread:

$$dc(K) = -\frac{d}{dK}c[K, \sigma(K)]. \quad (10)$$

Applying the chain rule gives the smile-adjusted digital price:

$$dc_{\text{smile}}(K) = dc_B(K) - \frac{\partial \sigma(K)}{\partial K} \mathcal{Vega}, \quad (11)$$

where

$$\mathcal{Vega} = B(t_0, t)F_0\varphi(d_1)\sqrt{\tau}. \quad (12)$$

The volatility slope is estimated by central finite differences on the cubic spline interpolant:

$$\left. \frac{\partial \sigma}{\partial K} \right|_{K=F_0} \approx \frac{\sigma(K + \varepsilon) - \sigma(K - \varepsilon)}{2\varepsilon}. \quad (13)$$

b. Results and Discussion

The following table summarizes the pricing results for the ATM digital call option on the Eurostoxx 50 index.

Method	Unit Price	Price (EUR)
Black	0.4352	217,617.50
Smile-Adjusted	0.5224	261,196.00
Difference	0.0872	43,578.50

Table 2: Digital option pricing results: Black vs. Smile-adjusted approach.

The smile-adjusted price exceeds the Black price by approximately **EUR 43,578** (~20%), a non-negligible discrepancy that arises from the **slope impact** of the implied volatility surface. As shown

in the following plot, the Eurostoxx 50 exhibits a pronounced **negative skew**: implied volatility decreases monotonically with the strike, yielding $\partial\sigma/\partial K < 0$ at the ATM point ($F_0 = 3763.97$, $\sigma_{\text{ATM}} = 23.72\%$). Since the smile correction term is $-(\partial\sigma/\partial K) \cdot \text{Vega}$, a negative slope produces a **positive correction**, pushing the smile price above the Black price. This result confirms that pricing digital options under a flat volatility assumption leads to a systematic **underestimation** of the fair value, as the Black model fails to capture the additional risk introduced by the discontinuity of the digital payoff in the presence of a skewed smile.

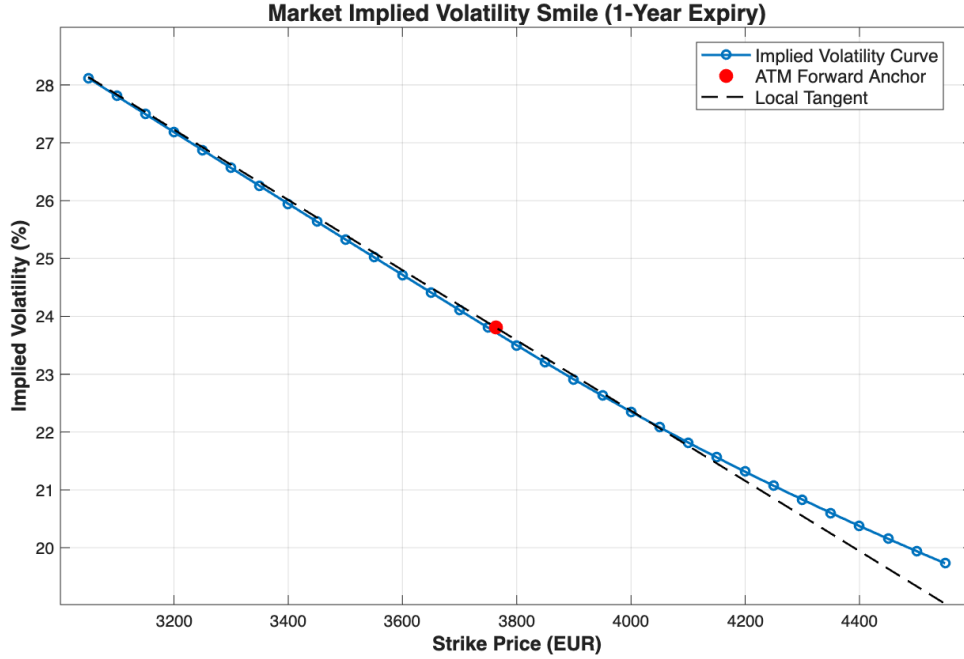


Figure 1: Market implied volatility smile (1-year expiry, Eurostoxx 50). The red dot indicates the ATM forward anchor at $F_0 = 3763.97$. The dashed line represents the local tangent, whose negative slope drives the smile correction.

3. Exercise 3: Pricing under a Given Characteristic Function

This exercise focuses on the pricing of European call options when the risk-neutral dynamics of the forward log-return are specified directly through a characteristic function.

In particular, we consider the following:

$$\phi(u) = \frac{1}{\left(1 - i \frac{u}{p^+}\right) \left(1 + i \frac{u}{p^-}\right)} e^{i\mu u}, \quad (14)$$

with $p^+ = 1.5$, $p^- = 0.9$, and μ chosen to satisfy the martingality condition.

The objective is to evaluate European call prices under this model and to compare four different computational approaches: numerical quadrature, Fast Fourier Transform (FFT), analytical evaluation by residues, and Monte Carlo simulation. The comparison is performed using the Eurostoxx 50 market inputs introduced in Exercise 2, namely a one-year maturity, the corresponding discount factor, and the ATM forward level F_0 . Prices are computed at three representative log-moneyness levels, $x = -5.223\%$, $x = 0\%$, and $x = 15\%$, in order to assess both numerical accuracy and robustness across strikes.

a. Pricing Methodologies

To evaluate the option price, four distinct mathematical and probabilistic approaches are implemented. The first three methods rely on resolving the fundamental **Lewis (2001) formula**, which expresses

the Call price in terms of log-moneyness $x = \ln(F_0/K)$:

$$C(x) = B(t_0, t) F_0 \left[1 - \frac{e^{-x/2}}{2\pi} \int_{-\infty}^{\infty} e^{-i\xi x} \frac{\phi(-\xi - \frac{i}{2})}{\xi^2 + \frac{1}{4}} d\xi \right] \quad (15)$$

where $B(t_0, t)$ is the discount factor. This integral is resolved via the following methodologies.

Quadrature (Numerical Integration)

The integral is solved pointwise for each specified moneyness level x using adaptive numerical quadrature (Gauss-Kronrod). This approach directly computes the area under the curve to a high specified tolerance, serving as a robust numerical benchmark.

Fast Fourier Transform (FFT)

To price options over a wide range of strikes simultaneously, the integral is discretized over a bounded grid of $N = 2^M$ points. By defining appropriate grid steps for the integration domain (dz) and the log-moneyness domain (dx) such that $dx \cdot dz = 2\pi/N$, the integral is cast into a Discrete Fourier Transform (DFT). This yields high computational efficiency, though it introduces truncation and discretization errors.

Analytical Evaluation (Residuals Technique)

Since the characteristic function $\phi(u)$ is rational, the Lewis integral can be evaluated analytically by extending the real-line integral to a contour integral in the complex plane on a semicircular path which incorporates the poles and applying Cauchy's Residue Theorem. The integrand can be written as

$$f(z) = \frac{e^{-iz(x+\mu)} e^{\mu/2} p^+ p^-}{(p^+ - \frac{1}{2} + iz) (p^- + \frac{1}{2} - iz) (z^2 + \frac{1}{4})}.$$

For $p^+ = 1.5, p^- = 0.9$, the poles are

$$z = \frac{i}{2}, \quad z = i$$

in the upper half-plane, and

$$z = -\frac{i}{2}, \quad z = -1.4i$$

in the lower half-plane.

The contour is closed in the upper half-plane when $x+\mu < 0$, and in the lower half-plane when $x+\mu \geq 0$. Since the integrand is a rational function multiplied by an exponential term, the contribution over the arc vanishes by Jordan's lemma. Therefore, the Lewis integral reduces to the sum of the residues inside the contour.

For $x + \mu < 0$,

$$I_{\text{UHP}}(x) = e^{x/2} - \frac{p^-}{(p^+ - 1)(p^+ + p^-)} e^{p^+(x+\mu)} e^{-x/2}.$$

With $p^+ = 1.5$ and $p^- = 0.9$,

$$I_{\text{UHP}}(x) = e^{x/2} - 0.75 e^{1.5(x+\mu)} e^{-x/2}.$$

For $x + \mu \geq 0$,

$$I_{\text{LHP}}(x) = e^{-x/2} - \frac{p^+}{(p^- + 1)(p^+ + p^-)} e^{-p^-(x+\mu)} e^{-x/2}.$$

With $p^+ = 1.5$ and $p^- = 0.9$,

$$I_{\text{LHP}}(x) = e^{-x/2} - 0.328947 e^{-0.9(x+\mu)} e^{-x/2}.$$

The call price is then

$$C = BF_0 \left[1 - e^{-x/2} I(x) \right],$$

where $I(x) = I_{\text{UHP}}(x)$ if $x + \mu < 0$, and $I(x) = I_{\text{LHP}}(x)$ otherwise. This closed-form formula provides the analytical benchmark for the numerical Lewis integral.

Monte Carlo Simulation

This methodology evaluates the option using a probabilistic Monte Carlo framework over N_{sim} iterations. It is implemented using two distinct simulation strategies, depending on whether the distribution can be analytically recognized:

Direct Simulation.

By closely analyzing the structure of the given characteristic function $\phi(u)$, one can recognize it as the moment-generating mechanism of a **Double Exponential (Laplace) distribution**. This mathematical insight allows the terminal log-return f_t to be simulated directly without requiring complex numerical inversion. The model can be represented as the difference of two independent exponential random variables. Let $Y^+ \sim \text{Exp}(p^+)$ and $Y^- \sim \text{Exp}(p^-)$. The terminal log-return is reconstructed as:

$$f_t = \mu + Y^+ - Y^- \quad (16)$$

The Call option price is subsequently calculated as the discounted arithmetic average of the simulated payoffs: $B(t_0, t) \cdot \max(F_0 e^{f_t} - K, 0)$.

Indirect Simulation (CDF Reconstruction via Digitals).

If the random variable X cannot be simulated directly from its characteristic function, we can simulate it indirectly by reconstructing its CDF on a grid using Fourier methods.

The CDF of X is recovered from the Lewis formula for digital options. In particular, if $\bar{D}_{\Delta u}^N(k)$ denotes the FFT approximation of the digital tail probability, then

$$\bar{P}_{\Delta u}^N(k) = 1 - \bar{D}_{\Delta u}^N(k).$$

We first select the numerical parameters α , Δu , and N , and compute $\bar{P}_{\Delta u}^N(k)$ on a grid of log-moneyness values k . Since a valid CDF must be increasing and bounded between 0 and 1, we cap and floor the computed values and retain only the interval where the reconstructed CDF is monotone increasing.

This gives a reduced grid $\bar{k}$ with endpoints $\bar{k}_{\min}$, $\bar{k}_{\max}$, and corresponding probability values $\bar{P}_{\min} = \bar{P}_{\Delta u}^N(\bar{k}_{\min})$, $\bar{P}_{\max} = \bar{P}_{\Delta u}^N(\bar{k}_{\max})$.

We then simulate independent uniform random variables

$$Z_j \sim \mathcal{U}[0, 1], \quad j = 1, \dots, N_{\text{sim}},$$

and obtain samples of X by numerical inversion of the reconstructed CDF:

$$X_j = (\bar{P}_{\Delta u}^N)^{-1}(Z_j).$$

In practice, we use **linear interpolation**. Hence, if

$$Z_j \in [\bar{P}_{\min}, \bar{P}_{\max}],$$

we compute

$$X_j = \text{interp}(\bar{P}_{\Delta u}^N(\bar{k}), \bar{k}, Z_j).$$

For values outside the interpolation range, we apply clamping:

$$\begin{aligned} Z_j > \bar{P}_{\max} &\Rightarrow X_j = \bar{k}_{\max}, \\ Z_j < \bar{P}_{\min} &\Rightarrow X_j = \bar{k}_{\min}. \end{aligned}$$

Finally, the call price is obtained by averaging the simulated discounted payoffs. This procedure provides an indirect simulation method based entirely on the CDF reconstructed from the Fourier-domain digital prices.

4. Exercise 4: Option Pricing under NTS Exponential Lévy Models

In the following exercises, we consider exponential Lévy models for the forward log-return in a one-step normal mean-variance mixture setting:

$$f_t = \log\left(\frac{F_t}{F_0}\right) = \sqrt{\Delta t} G \sigma g + \varphi \Delta t G + \mu, \quad g \sim \mathcal{N}(0, 1),$$

where G is a positive mixing random variable and μ is a deterministic correction chosen to enforce the martingale condition $\mathbb{E}[e^{f_t}] = 1$. The mixing variable is described through its Laplace transform

$$L[\omega] = \mathbb{E}[e^{-\omega G}],$$

with

$$\ln L[\omega] = \frac{\Delta t}{\kappa} \frac{1 - \alpha}{\alpha} \left[1 - \left(1 + \frac{\omega \kappa \sigma^2}{1 - \alpha} \right)^\alpha \right], \quad \alpha \in (0, 1).$$

We parametrize the drift component as $\mu = \left(\frac{1}{2} + \eta\right) \sigma^2$, where η controls the skewness. Under this parametrization, the characteristic function is obtained by evaluating the Laplace transform at $\omega = \frac{\xi^2 + i(1+2\eta)\xi}{2}$. Hence,

$$\phi_{f_t}(\xi) = e^{i\xi\mu} L\left[\frac{\xi^2 + i(1+2\eta)\xi}{2}\right].$$

This exercise addresses the pricing of European call options under a **Normal Inverse Gaussian (NIG)** model, corresponding to a normal mean-variance mixture with $\alpha = 1/2$, with parameters $\sigma = 20\%$, $\kappa = 1$, $\eta = 3$, and $\tau = 1$ year. Prices are computed for log-moneyness $x \in [-25\%, +25\%]$ on a grid with 1% steps, using the same forward F_0 , discount factor $B(t_0, t)$, and time to maturity as in Exercise 2. Three pricing methodologies are implemented and compared: FFT, quadrature, and Monte Carlo simulation via direct sampling of the NIG distribution.

a. Methodology

All pricing methods are based on the **Lewis formula**, as in Exercise 3. The characteristic exponent of the NIG model ($\alpha = 1/2$) is

$$\psi(u) = \frac{t}{\kappa} \left(1 - \sqrt{1 + \kappa \sigma^2 (u^2 + iu(1 + 2\eta))} \right), \quad (17)$$

The following pricing approaches are implemented.

- (a) **FFT.** The Lewis integral is computed on a uniform grid of $N = 2^M$ points via the Fast Fourier Transform. Prices at target moneyness values are recovered by linear interpolation.
- (b) **Quadrature.** The same integral is evaluated via adaptive numerical quadrature and used as the benchmark reference.
- (c) **Monte Carlo.** Log-returns are simulated directly under the NIG model using the Michael-Jones-Schucany algorithm. The IG parameters are linked to the model parameters through the Laplace transform of the mixing variable:

$$\mu_{IG} = t, \quad \lambda_{IG} = \frac{t^2}{\kappa}.$$

An alternative Monte Carlo estimator is also implemented through inverse transform sampling from the CDF reconstructed from digital prices.

- (d) **Facultative: $\alpha = 1/3$ model.** The same FFT and quadrature methods are applied to the $\alpha = 1/3$ normal mean-variance mixture, keeping the same parameters (σ, κ, η) .

b. Results and Discussion

(a)-(b) FFT and Quadrature.

Table 3 reports a selection of NIG call prices at representative moneyness values. Quadrature and FFT are in excellent agreement. The maximum absolute error is

$$\text{MAE}_{\max}^{FFT} = 3.51 \times 10^{-8},$$

while the mean absolute error is

$$\text{MAE}^{FFT} = 2.94 \times 10^{-9}.$$

x	K	Quadrature	FFT	Direct MC	MC from Digital
-25%	4833.03	27.7369	27.7369	27.7535	27.7171
0%	3763.97	290.2945	290.2945	290.5491	290.2098
+25%	2931.38	871.7199	871.7199	872.0178	871.6325

Table 3: NIG call prices (EUR) at selected moneyness values. The complete pricing grid is stored in `resultsEx4`.

The FFT sensitivity analysis shows that increasing the integration domain L reduces the truncation error. Among the tested values, $L = 80$ is selected by RMSE.

L	M	Max Abs. Error	RMSE Abs. Error
40	15	6.223347	3.135285
60	15	2.908356	1.403155
80	15	1.619605	0.787648

Table 4: NIG FFT sensitivity to the integration domain L .

(c) Monte Carlo.

Direct MC prices show deviations from quadrature consistent with simulation noise. The maximum absolute error is

$$\text{MAE}_{\max}^{MC} = 3.07 \times 10^{-1},$$

and the mean absolute error is

$$\text{MAE}^{MC} = 2.15 \times 10^{-1}.$$

The CDF reconstructed from digital prices is monotone increasing and spans almost the full probability range,

$$[9.48 \times 10^{-6}, 1].$$

The MC estimator based on inverse transform sampling from the reconstructed CDF gives smaller errors than direct MC, with

$$\text{MAE}_{\max}^{MC-dig} = 8.80 \times 10^{-2},$$

and

$$\text{MAE}^{MC-dig} = 6.58 \times 10^{-2}.$$

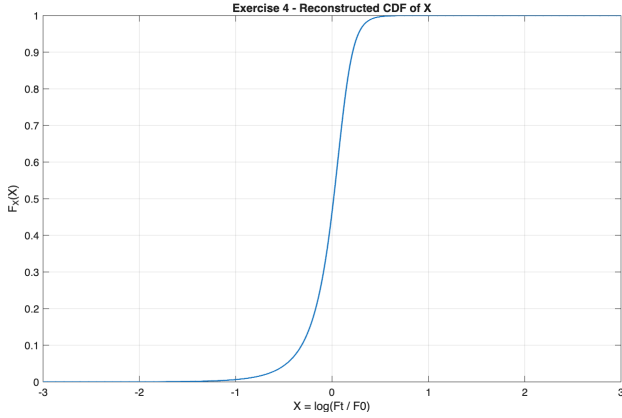


Figure 2: Reconstructed CDF of $X = \log(F_t/F_0)$ under the NIG model.

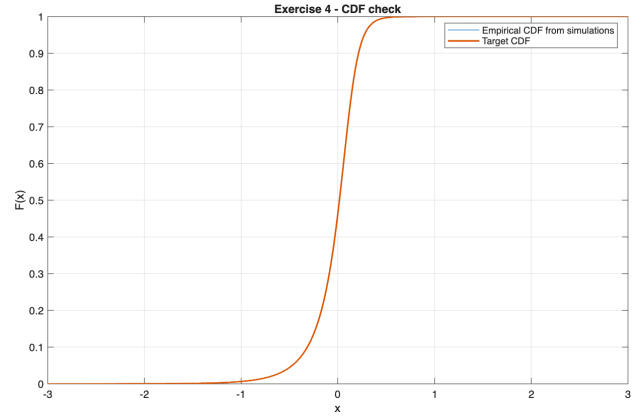


Figure 3: Empirical versus target CDF, validating inverse transform sampling.

NIG moment comparison.

The first raw moments obtained by finite-difference differentiation of the characteristic function are accurate for the first two moments. In particular,

$$|m_1^{num} - m_1^{ana}| = 1.24 \times 10^{-7}, \quad |m_2^{num} - m_2^{ana}| = 6.12 \times 10^{-8}.$$

The third moment is less precise but still acceptable for diagnostic purposes:

$$|m_3^{num} - m_3^{ana}| = 1.91 \times 10^{-4}.$$

However, the numerical estimate of m_4 is severely inaccurate for all tested values of h , with an absolute error of order

$$\mathcal{O}(10^0)$$

against the analytical value

$$m_4^{ana} = 7.7716.$$

This pathological behaviour arises because the fourth moment requires a fourth-order finite difference, which amplifies floating-point cancellation errors dramatically. For very small step sizes, such as $h = 10^{-4}$, the absolute error on m_4 drops to 5.55, but this apparent improvement is misleading: the error remains of order $\mathcal{O}(10^0)$ throughout all tested values of h , confirming that no choice of step size recovers a reliable estimate, and that machine precision cancellation dominates the stencil regardless of h .

(d) $\alpha = 1/3$ model.

For the $\alpha = 1/3$ normal mean-variance mixture, FFT remains consistent with quadrature but is less accurate than in the NIG case. The maximum absolute error is

$$\text{MAE}_{\max}^{NTS} = 1.23 \times 10^{-1},$$

while the mean absolute error is

$$\text{MAE}^{NTS} = 5.02 \times 10^{-2}.$$

Error Metric	NIG Model ($\alpha = 1/2$)	NTS Model ($\alpha = 1/3$)
Maximum Absolute Error	3.512372×10^{-8}	1.232386×10^{-1}
Mean Absolute Error	2.937680×10^{-9}	5.016026×10^{-2}

Table 5: Absolute pricing errors $|\text{FFT} - \text{Quadrature}|$ for the NIG and $\alpha = 1/3$ models.

The lower precision for $\alpha = 1/3$ is due to the slower Fourier-domain decay of the characteristic function. Since

$$|\phi(u)| \sim \exp(-c|u|^{2\alpha}),$$

the $\alpha = 1/3$ model decays as $\exp(-c|u|^{2/3})$, more slowly than the NIG case. Therefore, more Fourier tail mass remains far from the origin, making the FFT approximation more sensitive to truncation. This confirms that numerical grid parameters must be recalibrated according to the tail decay of the chosen Lévy model.

5. Case study: Volatility Surface Calibration

The objective of this exercise is to calibrate a **Normal Mean-Variance Mixture (NMVM)** model, specifically the **Normal Tempered Stable (NTS)** process ($\alpha = 2/3$), to the **EURO STOXX 50** implied volatility surface. The calibration is performed globally using constant weights. Finally, the model-implied volatility smile is plotted against the market data to assess the goodness of fit.

a. Methodology

Model call prices are computed using the quadrature approach. The calibration is performed by minimising a penalised Sum of Squared Errors between model and market prices, using constant weights ($w_i = 1$) across all available strikes:

$$\min_{\sigma, \kappa, \eta} \left\{ \sum_{i=1}^N (C_{\text{model}}(K_i; \sigma, \kappa, \eta) - C_{\text{mkt}}(K_i))^2 + \mathcal{P}(\sigma, \kappa, \eta) \right\}. \quad (18)$$

The penalty term is used to enforce the admissibility of the parameter vector during the unconstrained optimisation performed by `fminsearch`. In particular, the objective function assigns a large fixed value whenever the candidate parameters violate the basic positivity constraints

$$\sigma > 0, \quad \kappa > 0,$$

or the nonlinear admissibility condition

$$\eta \geq -\bar{\eta}(\sigma, \kappa),$$

where $\bar{\eta}(\sigma, \kappa)$ is the model-dependent lower bound for the skewness parameter. Equivalently, the penalty is defined as

$$\mathcal{P}(\sigma, \kappa, \eta) = \begin{cases} 0, & \sigma > 0, \kappa > 0, \eta \geq -\bar{\eta}(\sigma, \kappa), \\ \text{penaltyValue}, & \text{otherwise.} \end{cases}$$

In the implementation, if a parameter vector is inadmissible, the objective function immediately returns `penaltyValue`; otherwise, it computes the standard squared pricing error:

$$\text{Loss}(\sigma, \kappa, \eta) = \sum_{i=1}^N (C_{\text{model}}(K_i; \sigma, \kappa, \eta) - C_{\text{mkt}}(K_i))^2. \quad (19)$$

Thus, the optimisation remains numerically simple while preventing `fminsearch` from exploring regions where the NTS model is not well-defined.

b. Results and Discussion

The optimal parameter vector $(\hat{\sigma}, \hat{\kappa}, \hat{\eta})$ obtained from the calibration is reported in Table 6.

Table 6: Calibrated NTS model parameters ($\alpha = 2/3$).

Parameter	Symbol	Value
Volatility level	$\hat{\sigma}$	0.1374
Vol-of-vol	$\hat{\kappa}$	0.2564
Skewness	$\hat{\eta}$	25.2061
Objective value	$\hat{f}$	5.8248

Figure 4 compares the market implied volatility smile with the NTS model-implied one, plotted against the log-moneyness $x = \log(F_0/K)$.

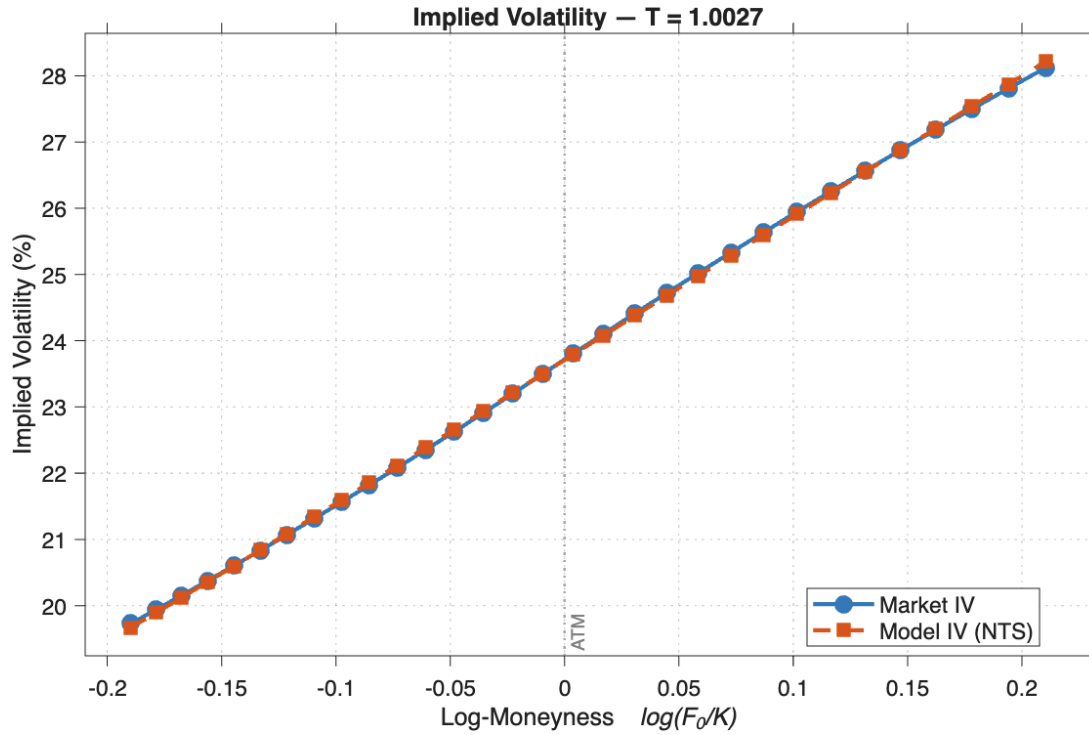


Figure 4: Implied volatility smile: market (blue circles) vs. NTS model (orange squares). The vertical dashed line marks the ATM point ($x = 0$).

The graphical output confirms an excellent fit. The negative slope of the smile (higher implied volatility for $x > 0$, i.e., OTM puts) accurately reflects the **negative skew** characteristic of major equity indices like the EURO STOXX 50. This phenomenon is driven by the structural market demand for downside protection against market crashes.